

Unit 8, Lesson 3: Exponential Expressions

Benchmark

MA.7.NSO.1.1 Know and apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational number bases.

Learning Targets

- ☐ I can evaluate numeric expressions with the same base using the laws of exponents.
- ☐ I can generate equivalent numeric expressions by applying the laws of exponents.

8.3.1 Warm-Up: Make an Expression

1. Create an expression with the largest possible value by filling in the blue boxes with the digits 1 through 5. You can only use each digit one time.

$$\square + \square \times \square$$

8.3.2 Collaboration: Applying More Than One Exponent Law

An expression can be rewritten as an equivalent expression by applying one or more exponent laws.

Winston and Yvette were assigned $\left(\frac{2^6}{2^4}\right)^3$ to rewrite as an equivalent expression using multiple exponent laws. Their teacher told them to first make a plan and then apply the laws to the expression. Complete Yvette’s work by following her plan.

Winston	Yvette
I will first apply the quotient of powers exponent law and then apply the power of a power exponent law.	I will first apply the power of a quotient exponent law, then the power of a power exponent law, and finally the quotient of powers exponent law.
$\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3 = 2^6 = 64$	$\left(\frac{2^6}{2^4}\right)^3 = \frac{(2^6)^3}{(2^4)^3} =$

1. Explain which method you prefer.
2. Ask your partner which method they prefer. Write their explanation here.

8.3.3 Guided Practice: Writing Equivalent Expressions to Evaluate

1. Two students evaluated the expression $\frac{4^5 \cdot 4^0}{4^2}$ and found its value.

Caroline's Work	Jamir's Work
$\frac{4^5 \cdot 4^0}{4^2}$	$\frac{4^5 \cdot 4^0}{4^2}$
$\frac{4^{(5+0)}}{4^2}$	$\frac{4^5 \cdot 1}{4^2}$
$\frac{4^5}{4^2}$	$\frac{4^5}{4^2}$
$4^{(5-2)}$	$4^{(5-2)}$
4^3	4^3
64	64

Complete the statements explaining each student's work by using the phrases from the word bank.

zero

identity

power of a power

product of powers

power of a product

quotient of powers

power of a quotient

- a. Caroline used the _____ exponent law first by rewriting $4^5 \cdot 4^0$ as 4^5 by adding the exponents. Next, she used the _____ exponent law by subtracting the exponents 5 and 2. Then, Caroline found the value of the exponential expression.

zero
identity
power of a power
product of powers
power of a product
quotient of powers
power of a quotient

- b. Jamir used the _____ exponent law first by writing 4^0 as 1. Then, he multiplied 4^5 by 1. Next, he used the _____ exponent law by subtracting the exponents 5 and 2. Finally, he found the value of the exponential expression.

- c. The students used the laws differently, but each expression they wrote is

- ☐ less than
☐ equivalent to
☐ greater than

$$\frac{4^5 \cdot 4^0}{4^2}$$

2. Ruth was given the following problem to evaluate.

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4}$$

- a. Why would rewriting the following expression using laws of exponents be useful to Ruth before evaluating the expression?
- b. Rewrite the expression using as many laws of exponents as possible. Then, evaluate the expression.

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4}$$

8.3.4 Activity: Roundtable

In teams of three, assign a number to each team member. Each student completes the **first** step of their problem and writes the exponent law used. Then, pass your problem to the student on your right to check your work and write the next step to the problem. For each new step, continue passing the problems to the student on your right until all three problems are evaluated.

Team member 1: _____	
$(3^0 \cdot 3^2)^3$	Exponent law used
Team member 2: _____	
$\frac{(0.5^3)^2}{(0.5^1)^2}$	Exponent law used

Team member 3: _____	
$\left(\frac{4^0}{4^2} \cdot 4^4\right)^3$	Exponent law used

8.3.5 Your Turn!

1. Use the laws of exponents to evaluate the expressions. Record the law used at each step.

Equivalent Expressions	Laws of Exponents
$\left(\frac{2.5^5}{2.5^3}\right)^1$	
$\left[\left(\frac{2}{3}\right)^0 \cdot \left(\frac{2}{3}\right)^2\right]^3$	
$\frac{0.5^3}{2.5^3} \cdot \left(\frac{0.5}{2.5}\right)^2$	

2. Evaluate each expression showing each step of your work. Use a calculator to verify your solutions.

a. $\left(\frac{0.4^2}{0.4^1}\right)^2$

b. $\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$

c. $\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$

d. $0.1^2 \cdot (0.1^1)^2$

e. $\frac{2.1}{3 \cdot 2^0} \cdot 2.1^2$

8.3.6 Wrap-Up: Ticket Out the Door

1. Use the laws of exponents to evaluate $\frac{0.2^3}{2^3} \cdot 0.1^2$. Show your work.

2. Select all of the expressions equivalent to $\frac{0.2^3}{2^3} \cdot 0.1^2$.

☐ 0.1^5

☐ $0.2^0 \cdot 0.1^2$

☐ $0.1^3 \cdot 0.1^2$

☐ $\frac{0.2^5}{0.2^3}$

☐ $\left(\frac{0.2}{2}\right)^3 \cdot 0.1^2$

☐ 0.2^5

